

Appendix A - South-west Corner Marine Park (Western arm) Natural Values Report

South-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

Date	Campaign name	Method	Areas	Samples	Habitat	Fish	Length
					Count	Count	
Jun 2020	2020-06 south-west stereo-BRUVs	BRUV	NPZ	42	40	39	33
Oct-Nov 2020	2020-10 south-west stereo-BRUVs	BRUV	Coastal waters, NPZ, SPZ	275	273	272	244
Oct-Nov 2020	2020-10 south-west BOSS	BOSS	Coastal waters, NPZ, SPZ	262	262	-	-
Mar 2021	2021-03 West-Coast BOSS	BOSS	Coastal waters, NPZ	154	154	-	-
Mar 2023	2023-03 Augusta stereo-BOSS	BOSS	Coastal waters, NPZ, SPZ	73	70	-	-
Mar 2023	2023-03 SwC BOSS	BOSS	NPZ	88	88	-	-
Mar 2023	2023-03 SwC stereo-BRUVs	BRUV	NPZ	32	32	32	32
Oct 2024	2024-10 SWC stereo-BRUVs	BRUV	Coastal waters, NPZ, SPZ	190	190	188	178
Nov 2024	2024-11 SwC stereo-BOSS	BOSS	Coastal waters, MUZ, NPZ, SPZ	137	117	-	-
Mar-Apr 2025	2025-04 South-west Corner BOSS	BOSS	NPZ, SPZ	210	210	-	-
Apr 2025	2025-04 South-west-corner stereo-BRUVs	BRUV	NPZ, SPZ	150	148	143	137

Benthic ecosystem extent

Observations

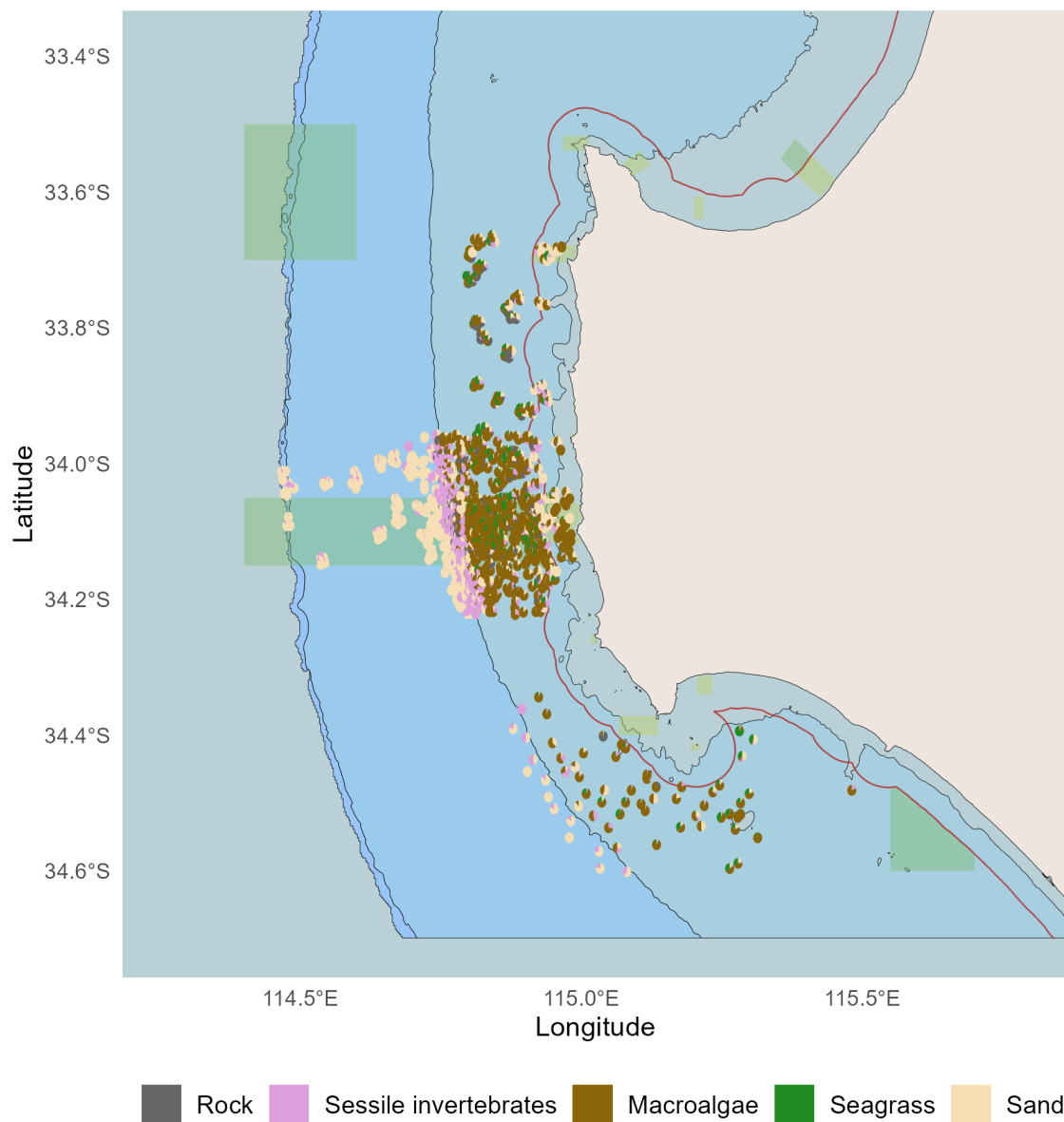


Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

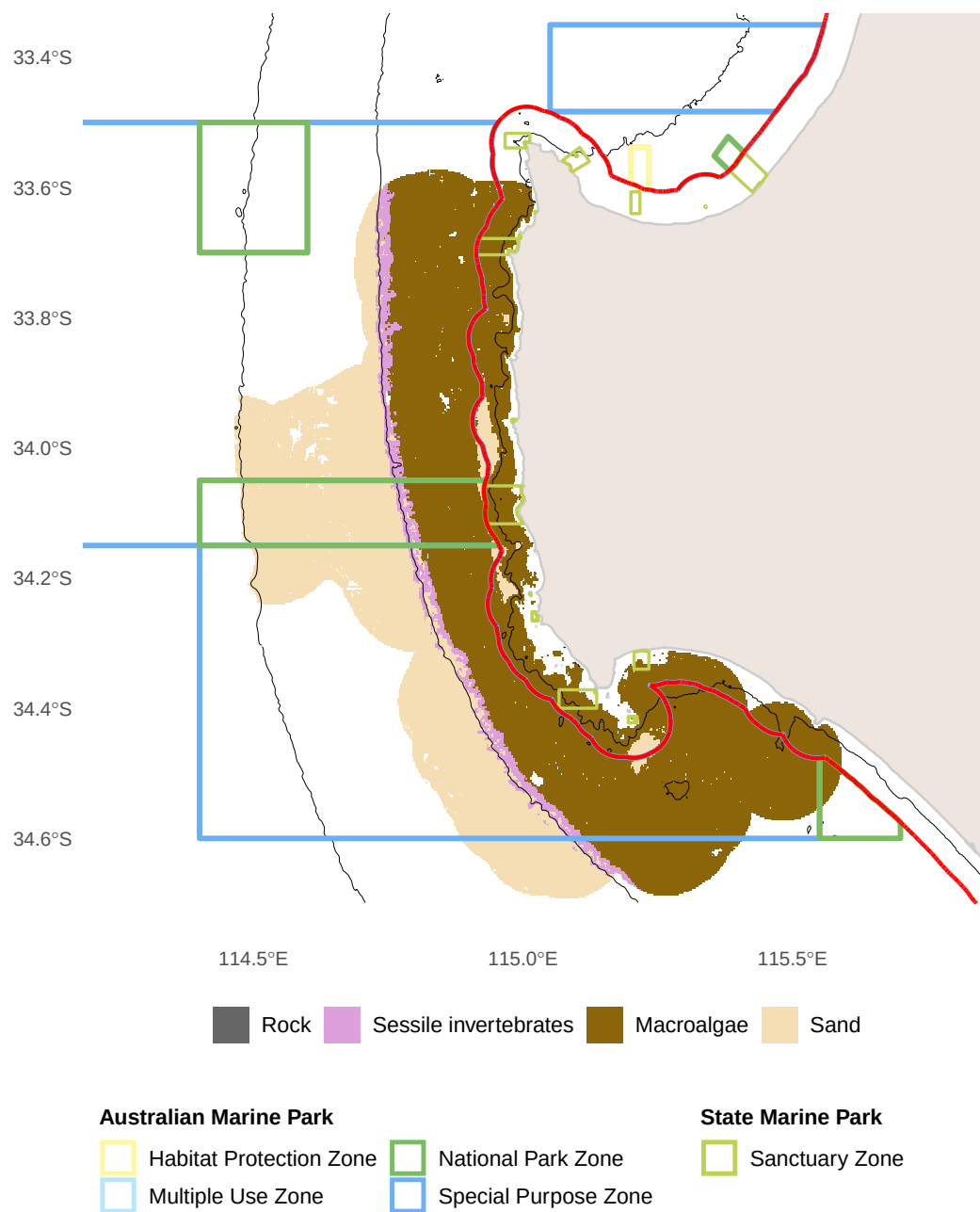


Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

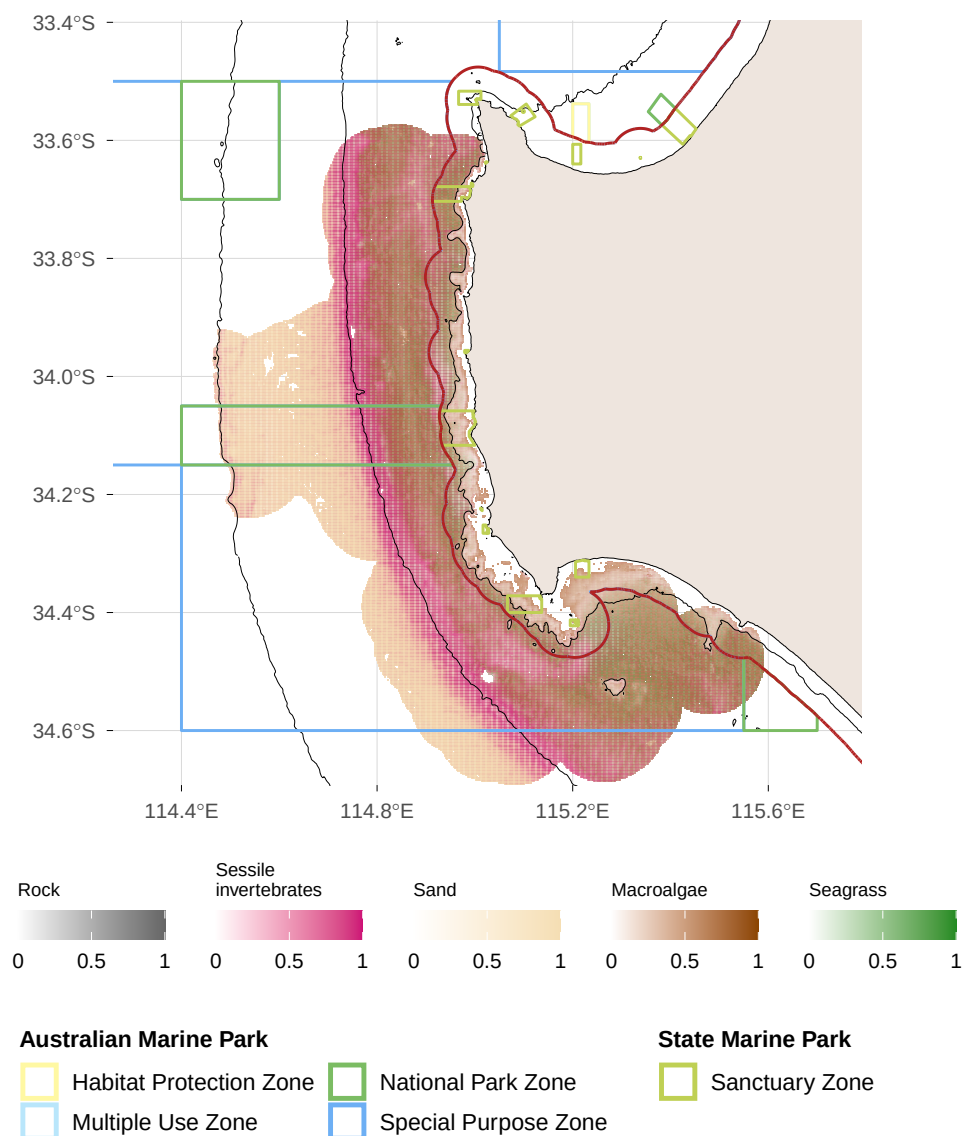


Figure 3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

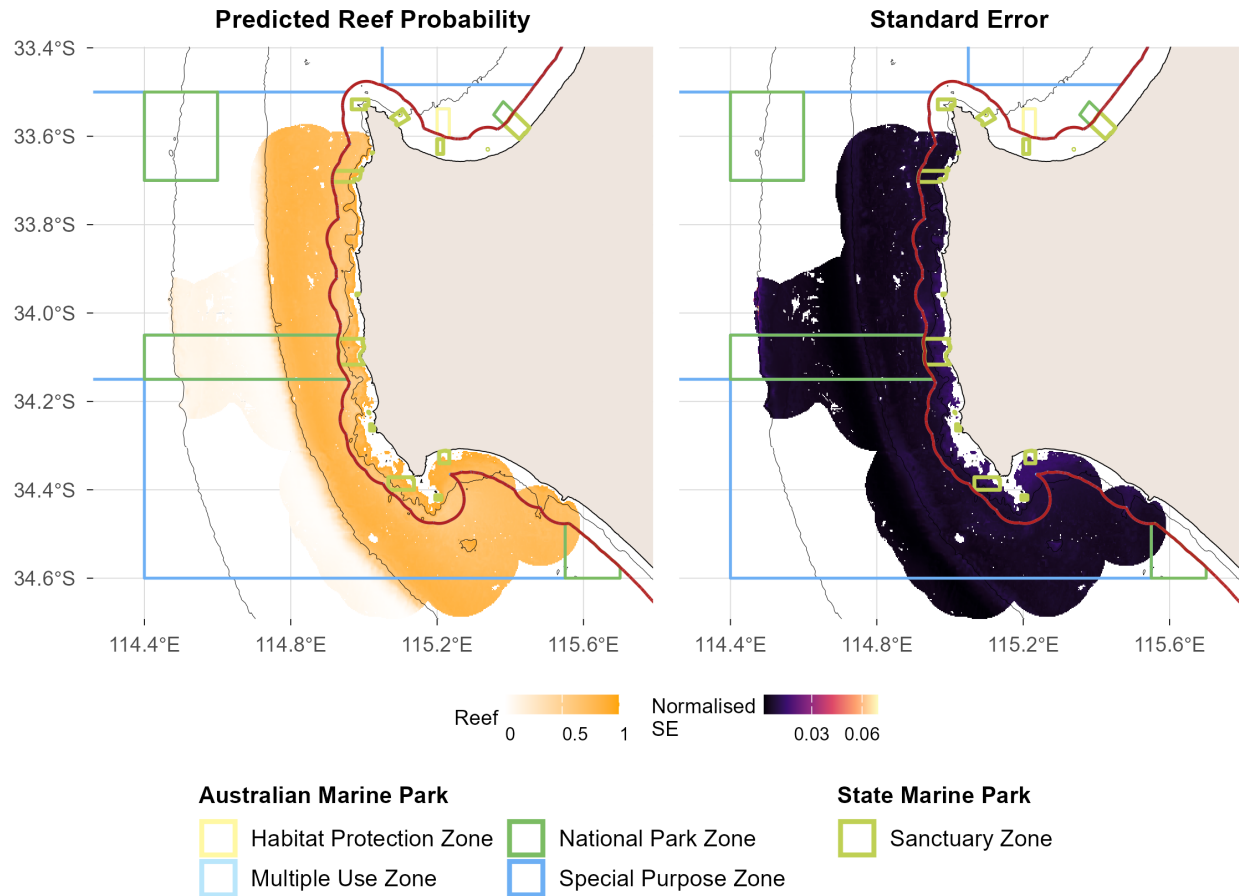


Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

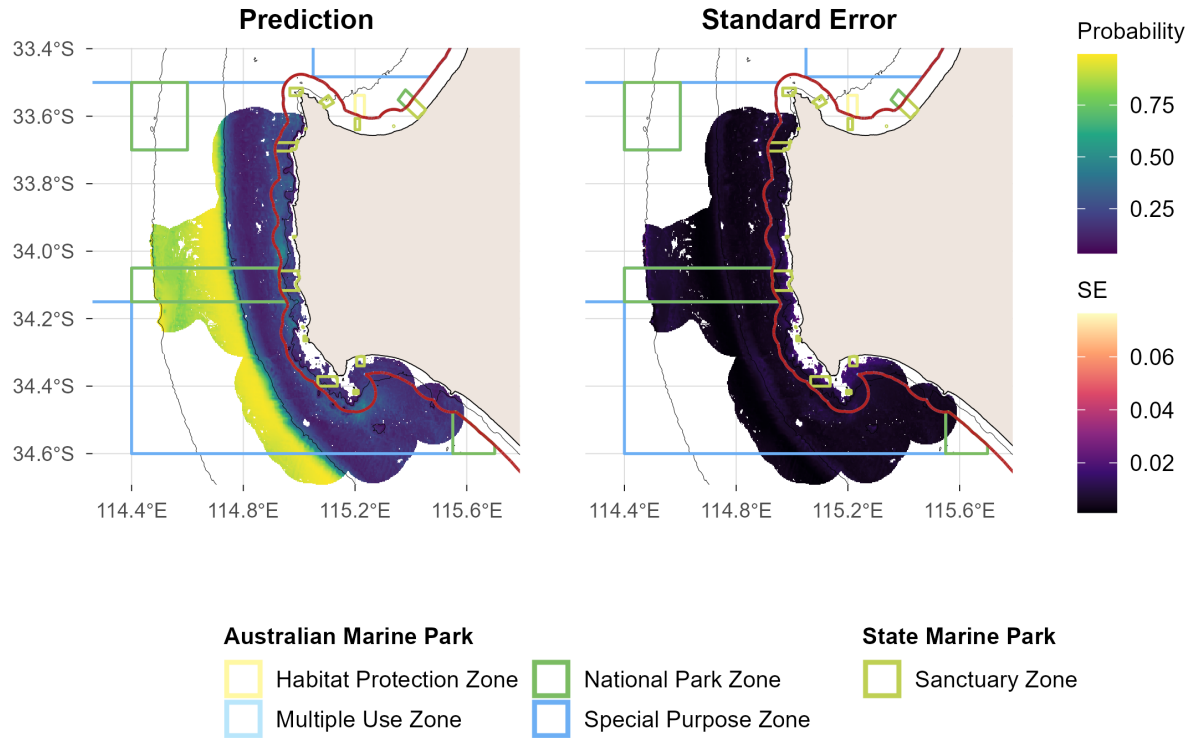


Figure 5.1: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

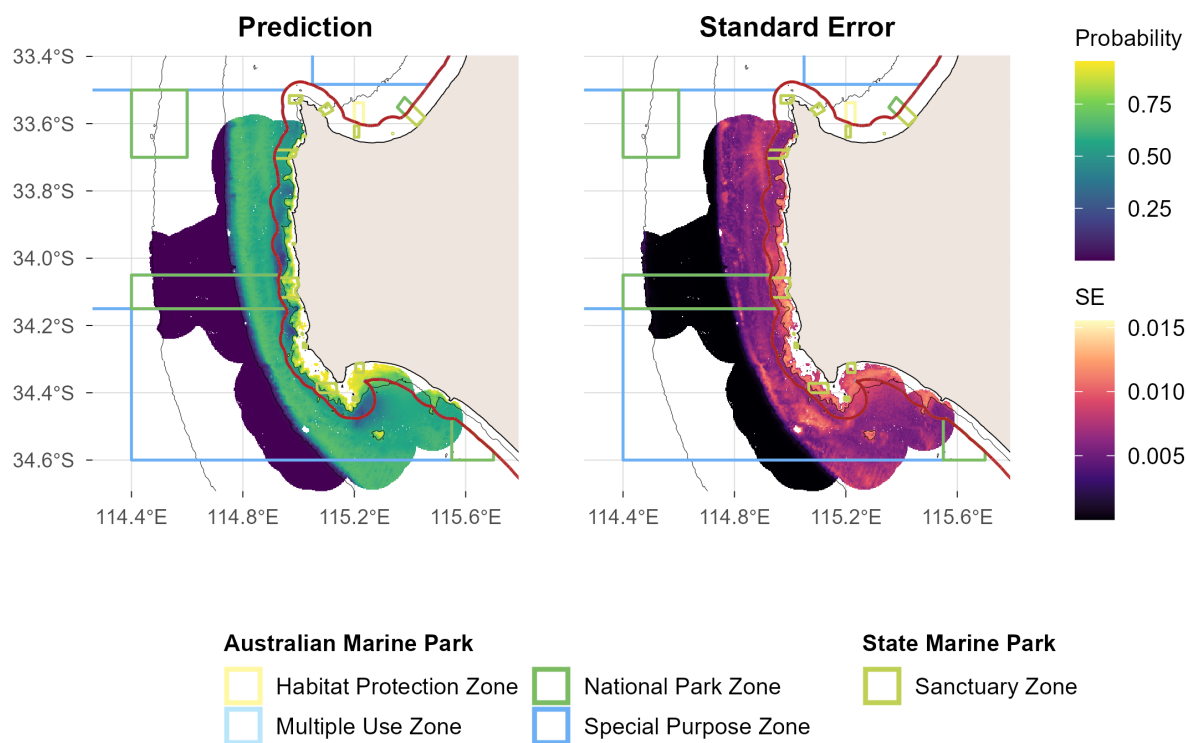


Figure 6.1: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Seagrass

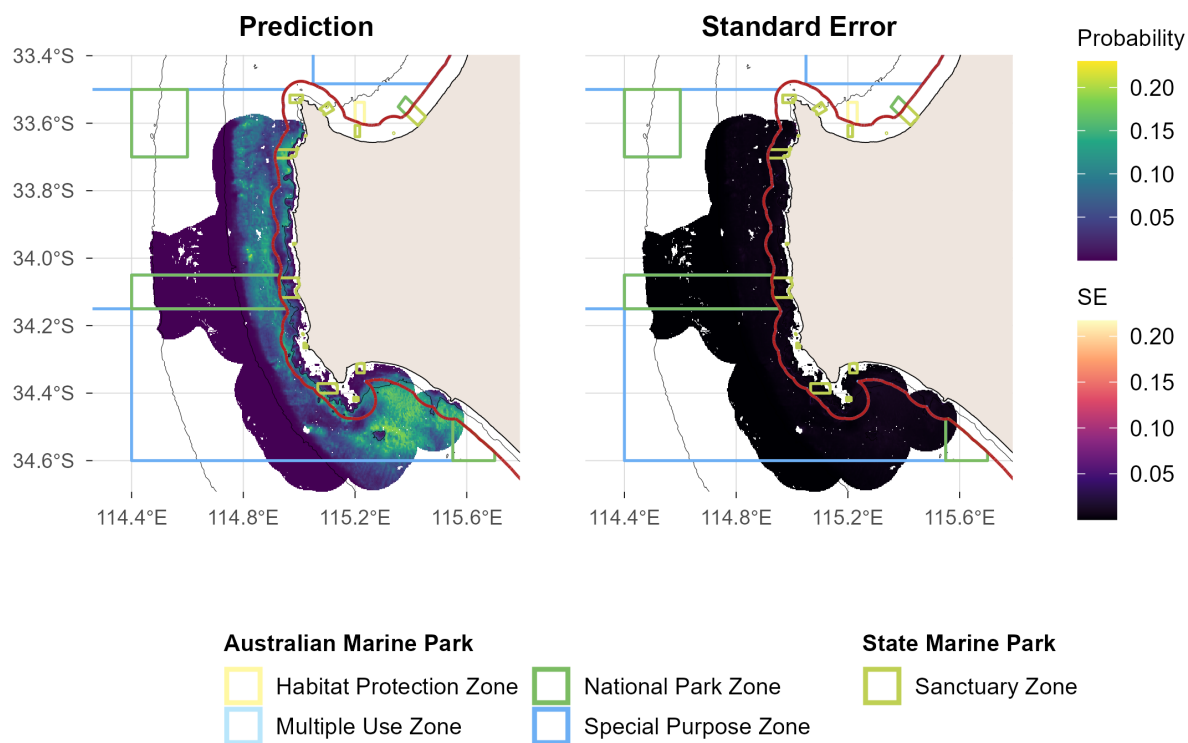


Figure 7.1: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

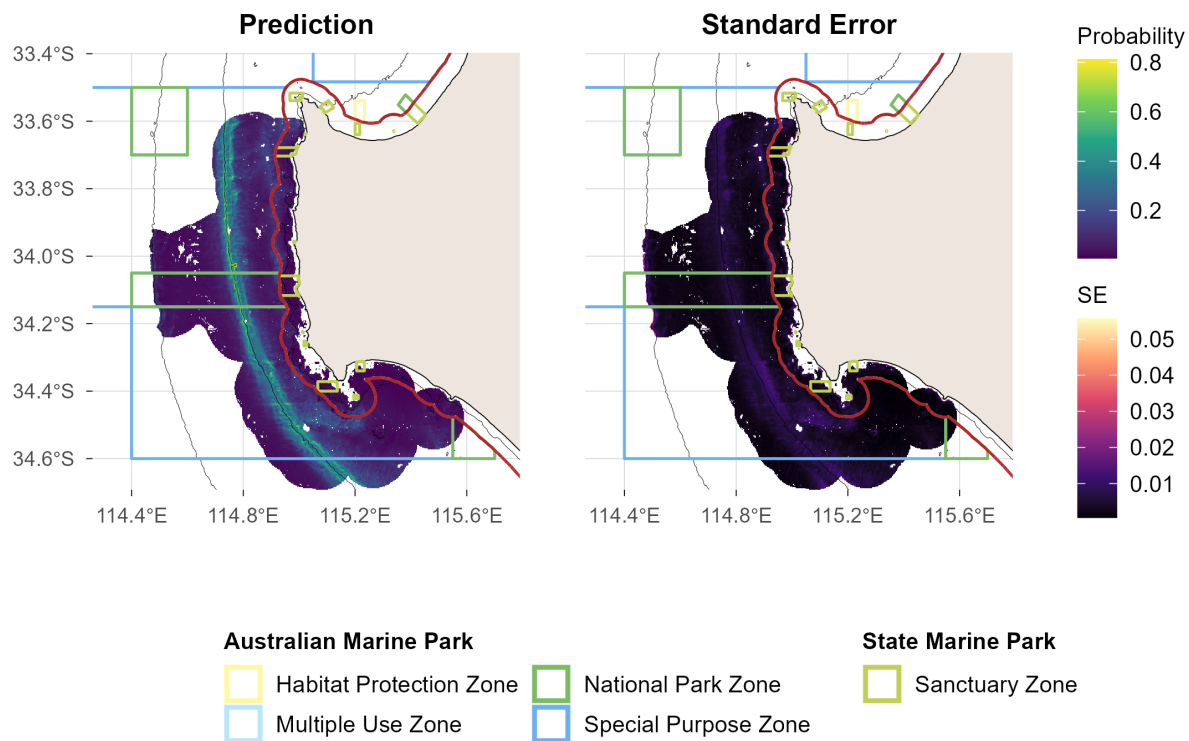


Figure 8.1: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Rock

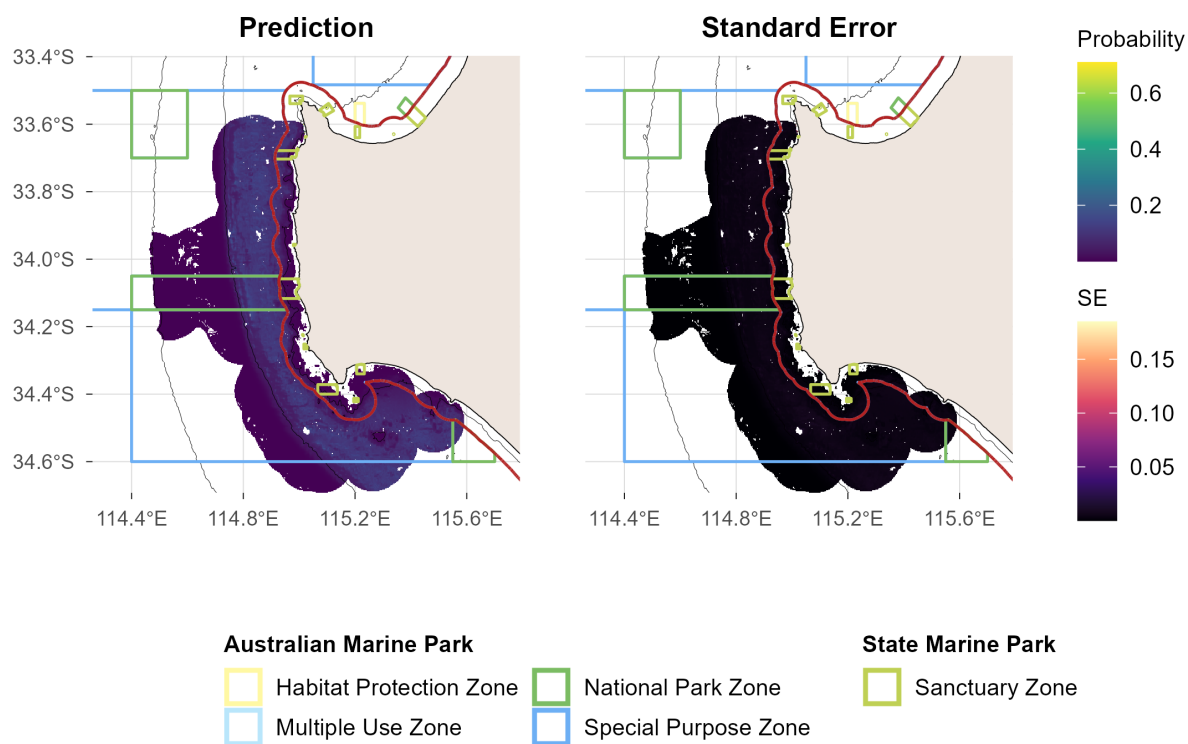
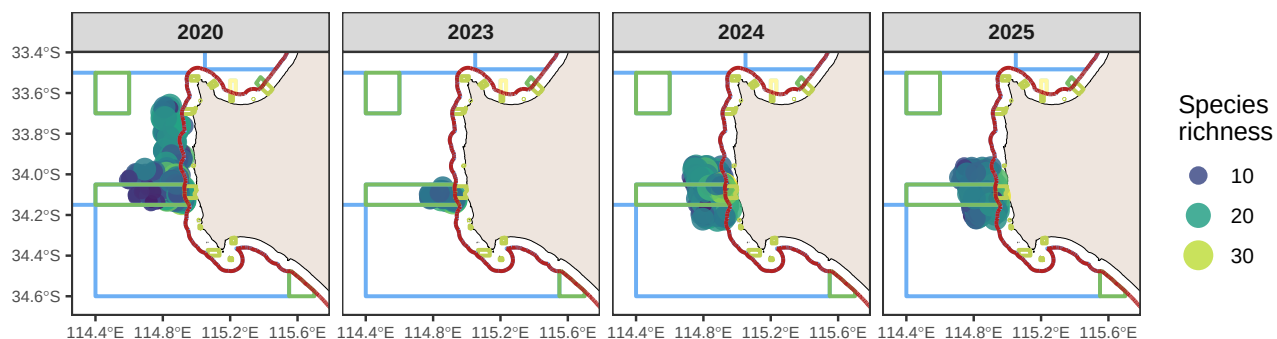


Figure 9.1: Rock: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

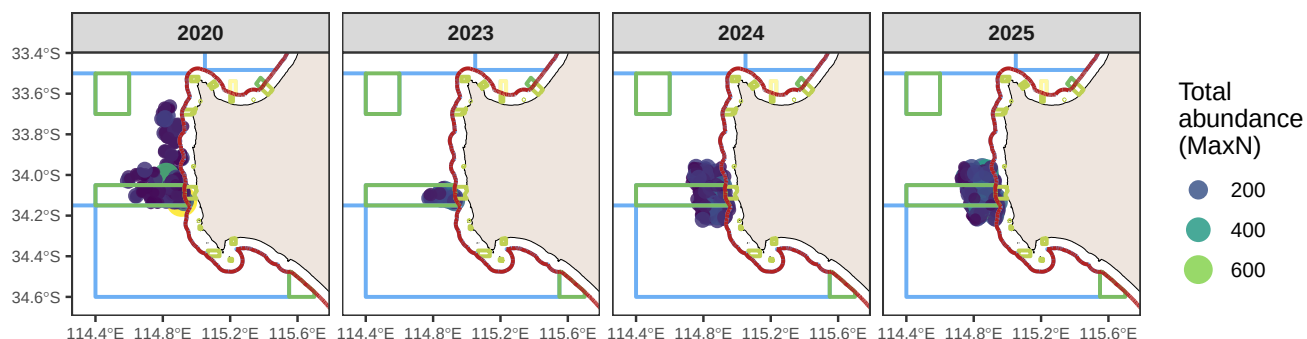
Benchmarks of demersal fish natural values

Survey effort

a)



b)



Australian Marine Park

Habitat Protection Zone
Multiple Use Zone

National Park Zone
Special Purpose Zone

State Marine Park

Sanctuary Zone

Figure 10: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

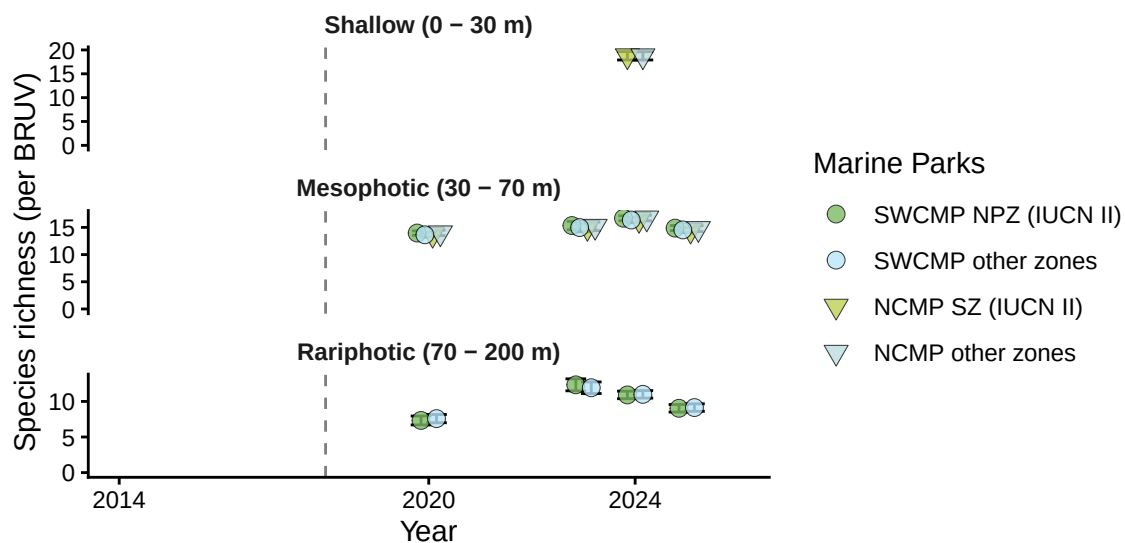


Figure 11.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

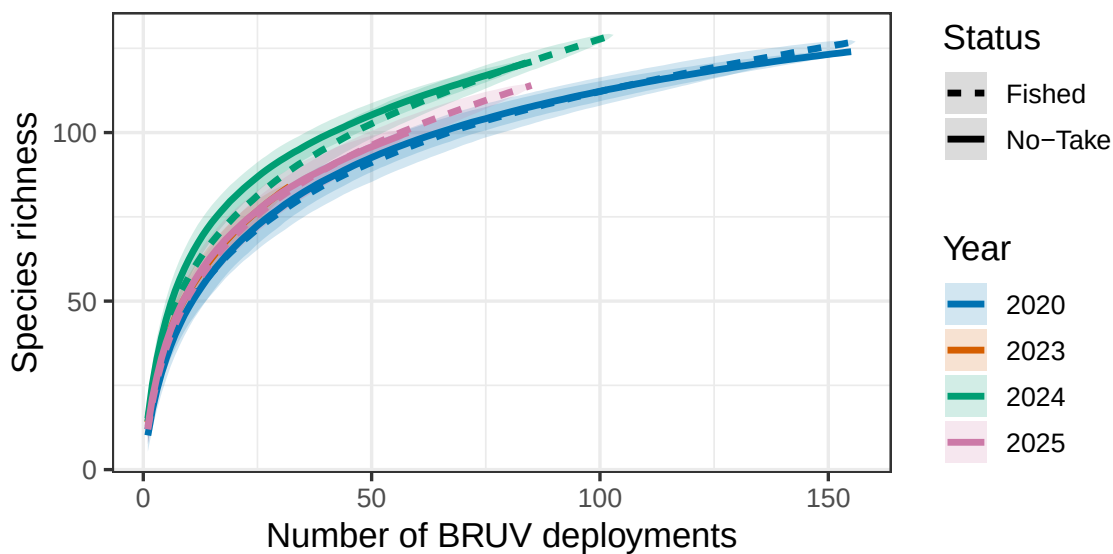


Figure 11.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Species richness

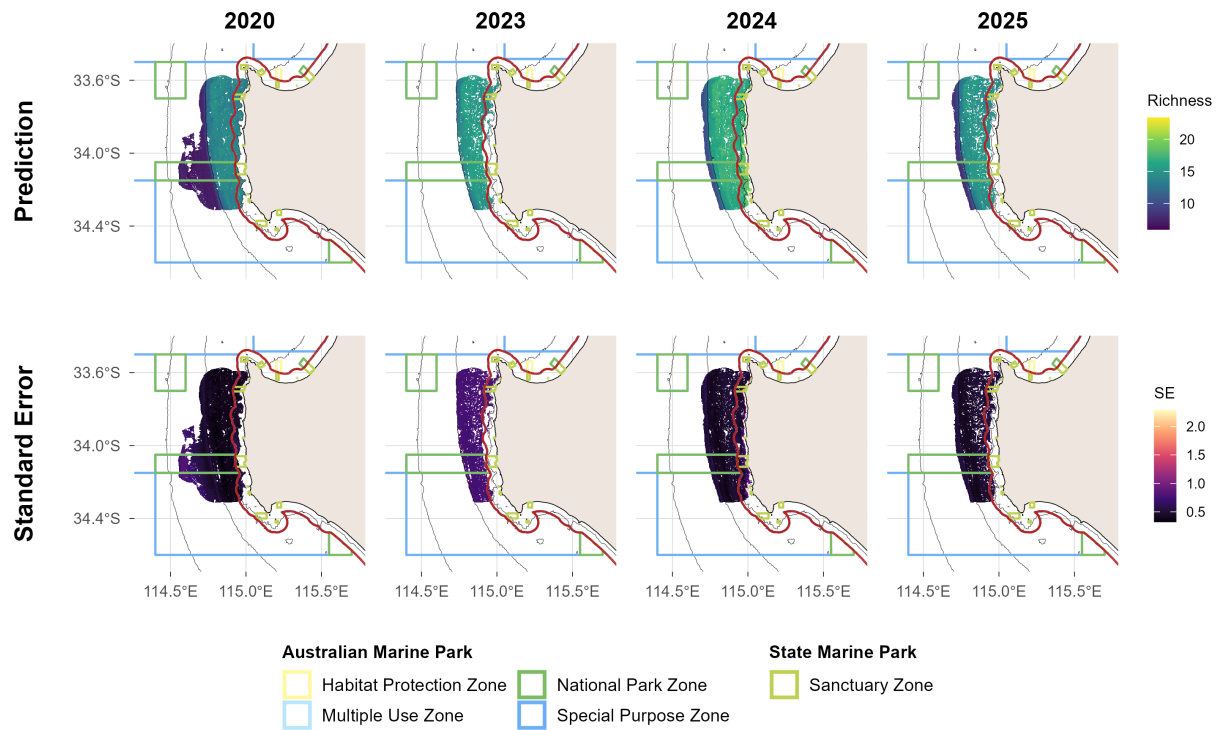


Figure 11.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

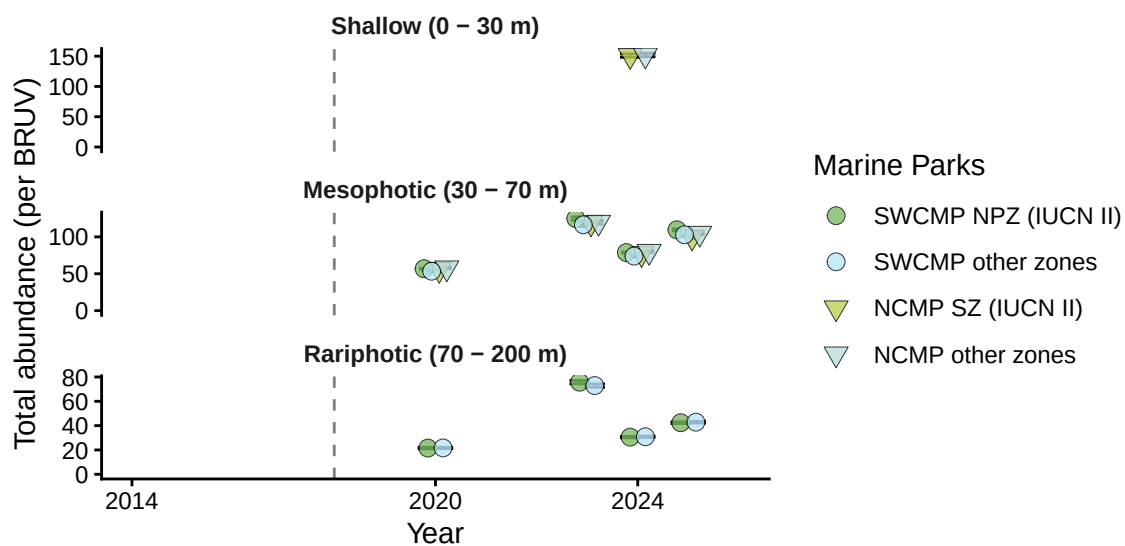


Figure 12.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

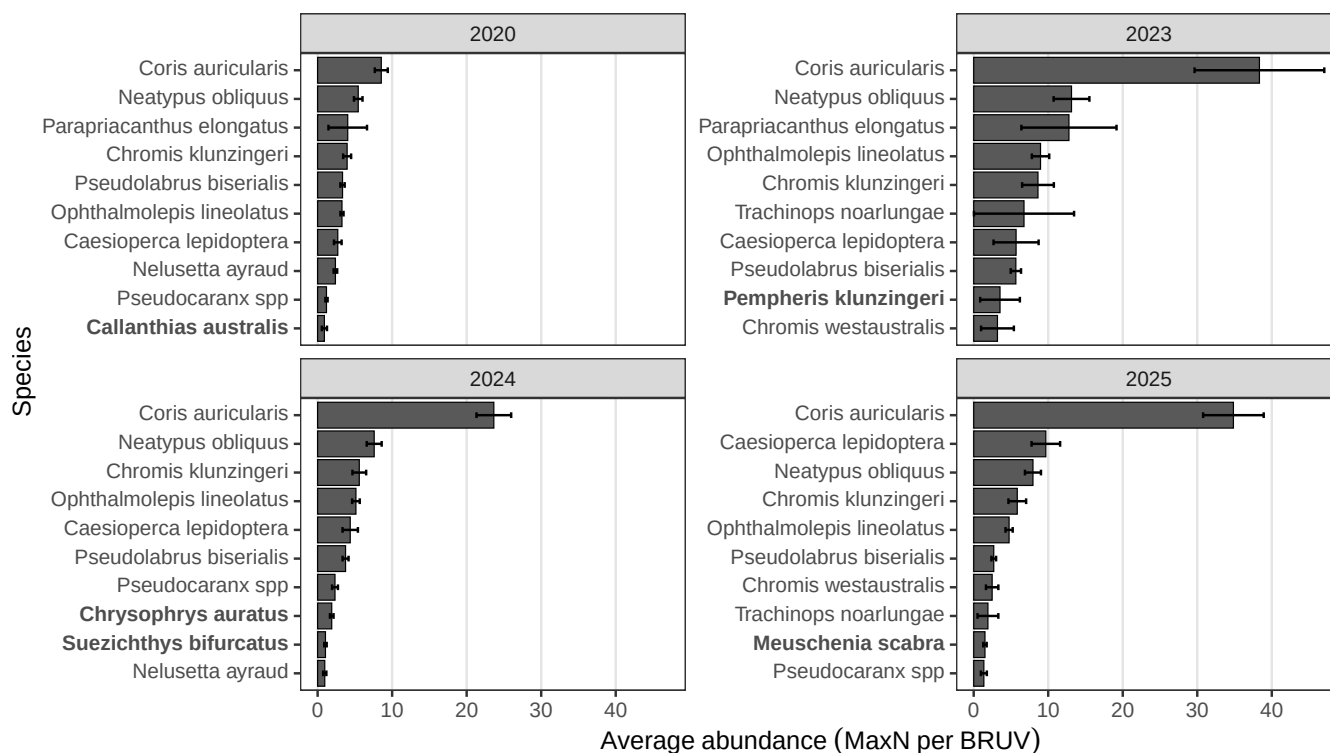


Figure 12.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Total abundance

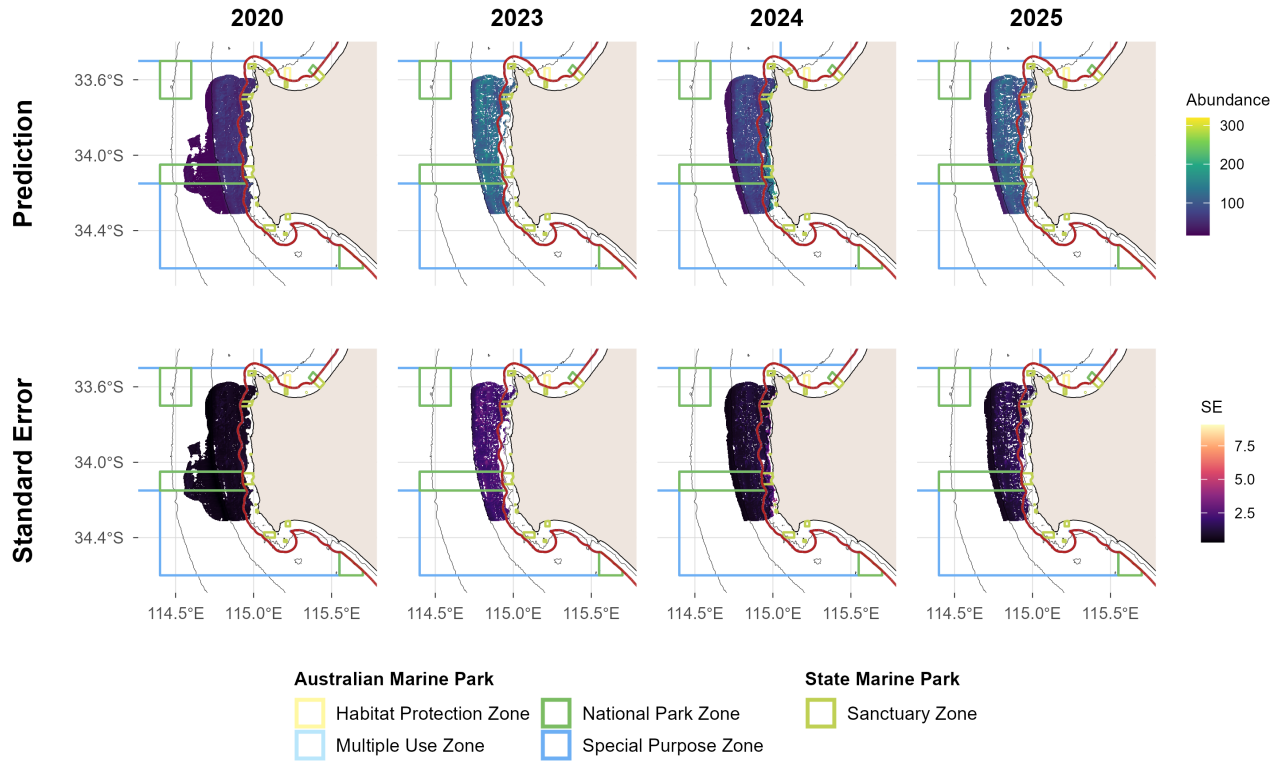


Figure 12.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

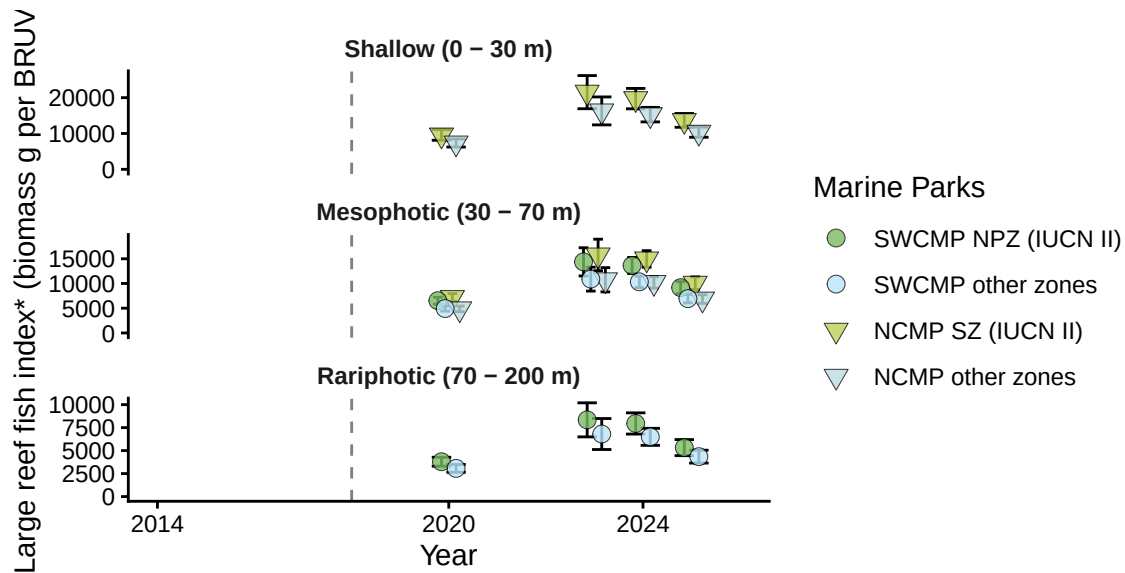


Figure 13.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

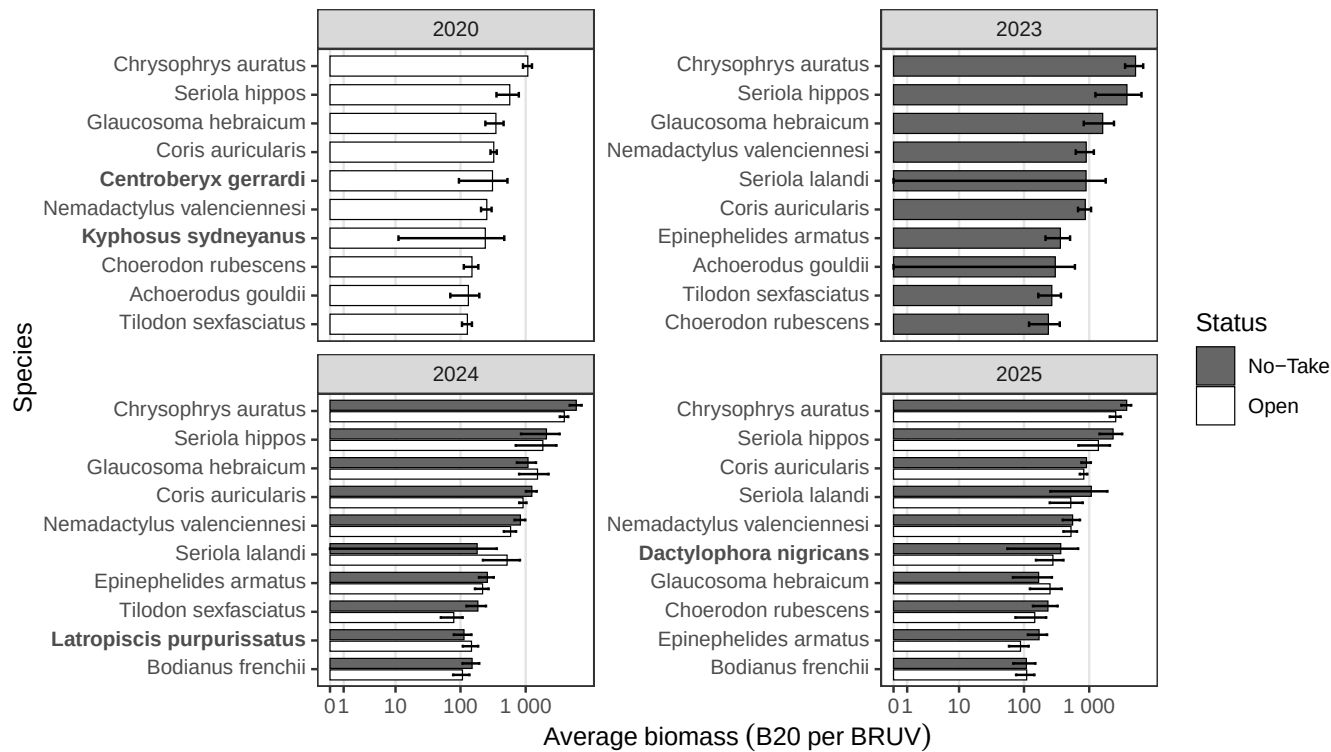


Figure 13.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Large Reef Fish Index*

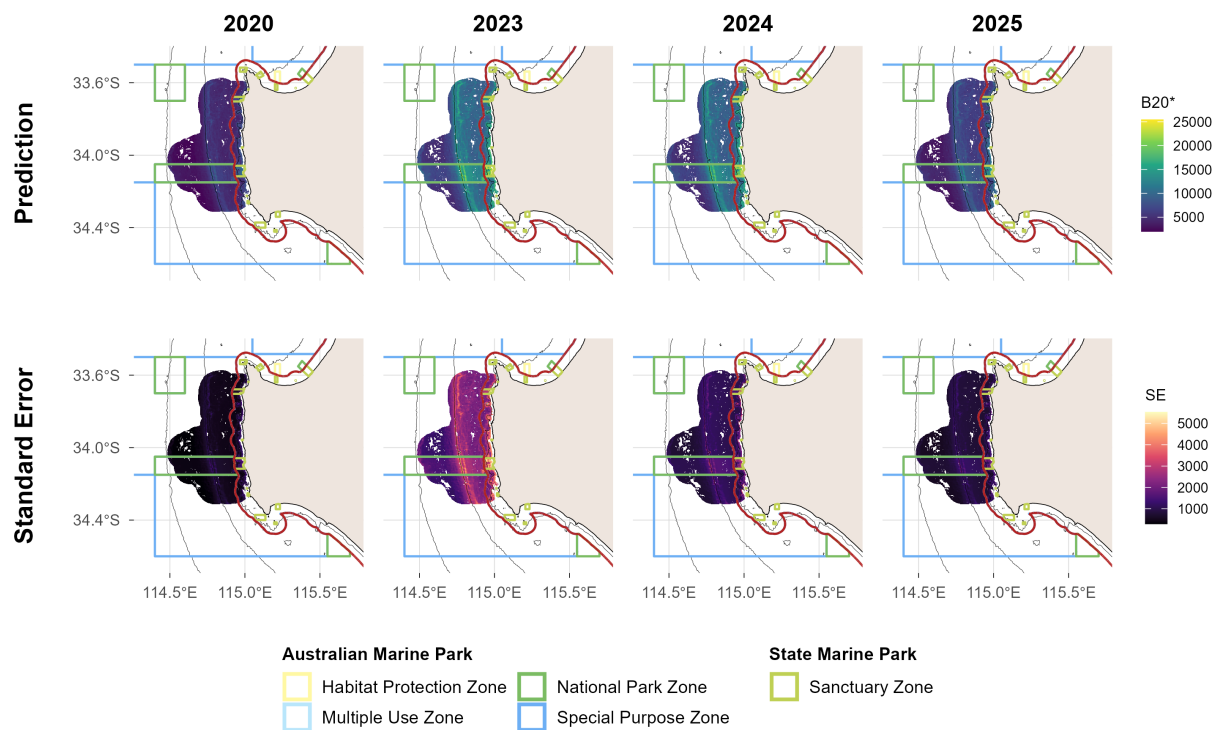


Figure 13.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

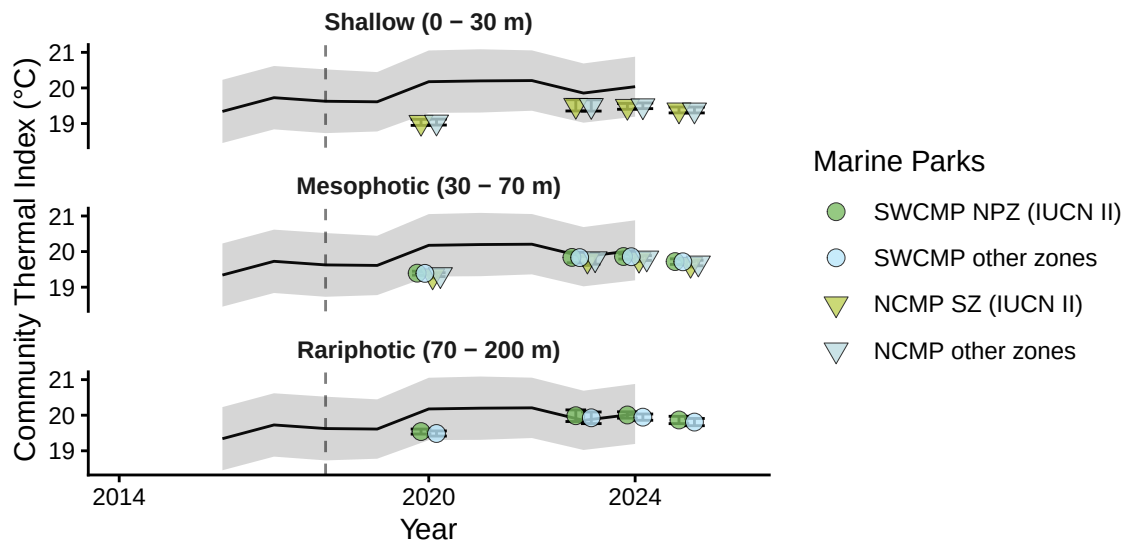


Figure 14.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

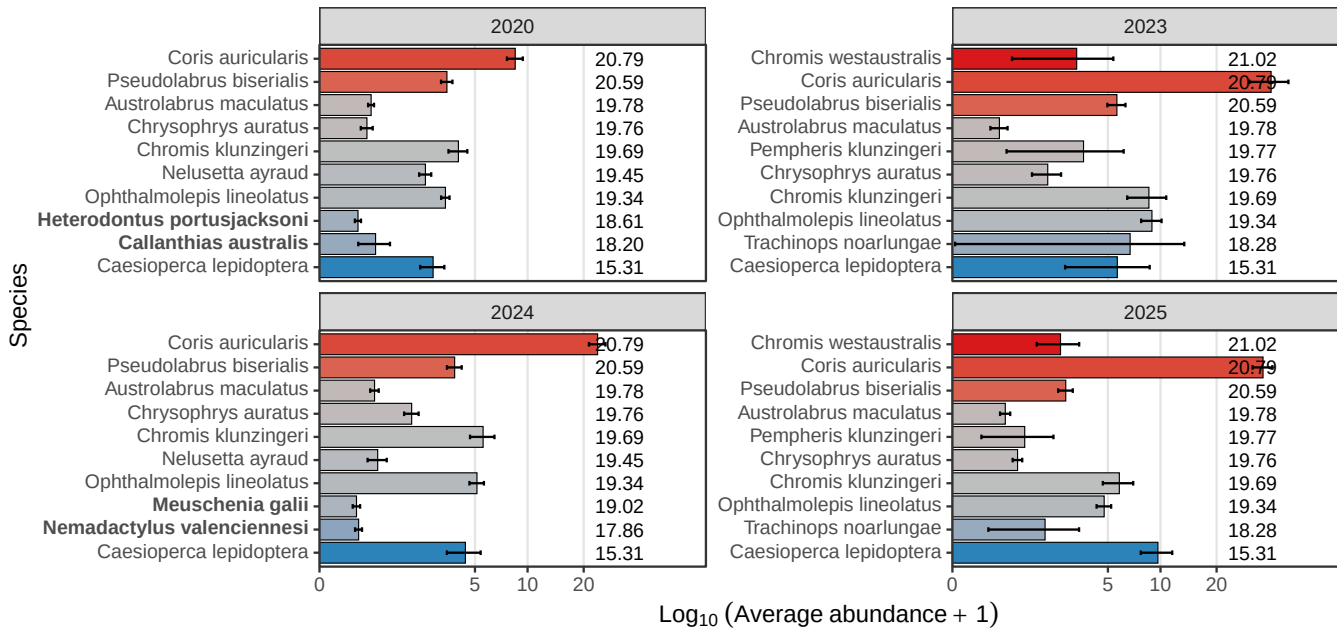


Figure 14.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Reef fish thermal index

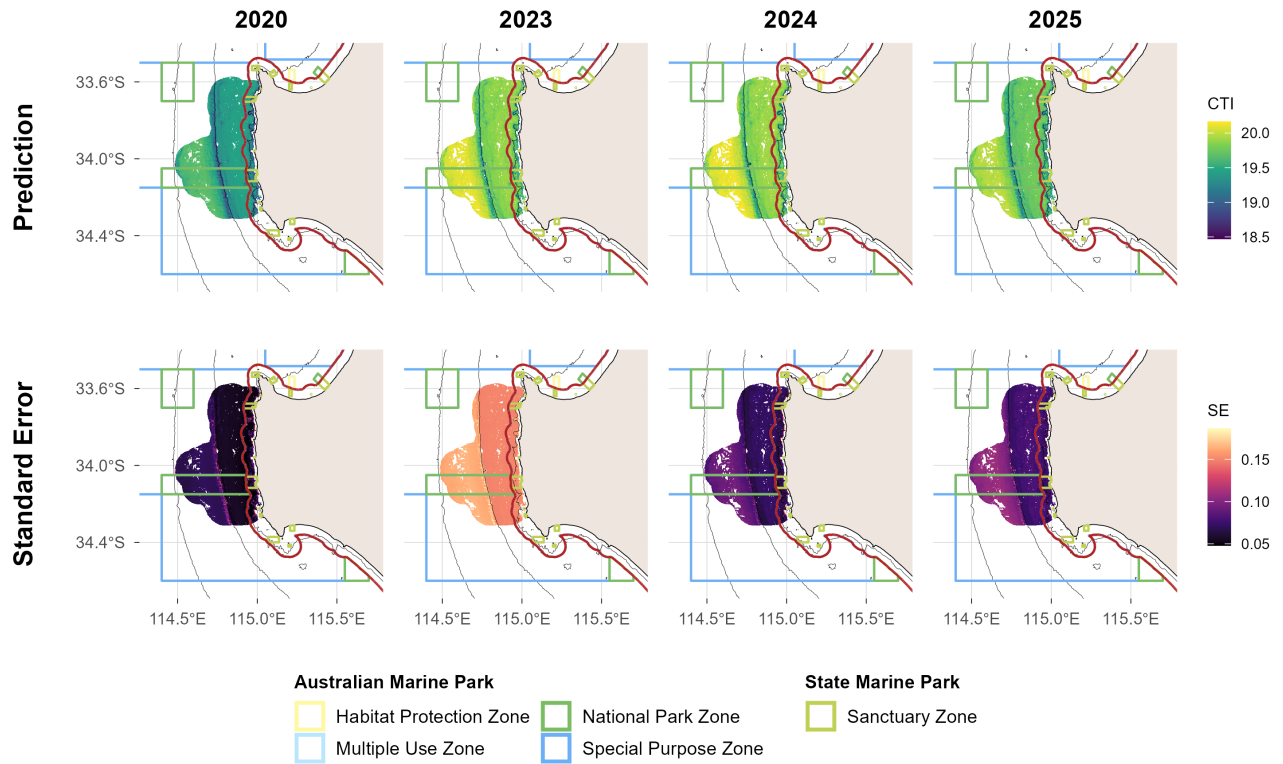


Figure 14.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.